

known – most projectors are less than 2 kg in weight and have above 2500 lumens of brightness and a smaller footprint than laptops. While this may not sound as ‘impressive’ as a mobile phone projector, the fact is that these mainstream projectors are the ones that meet the widest range of user applications without constraining the presenters. For specialised applications, there are higher-brightness (3000-5000-lumen) projectors available, but these come in a much larger and heavier package,” says Abhimanyu Gupta, director, Actis Technologies.

Every manufacturer has managed to reduce the size of its projectors, through one or the other technological means. Epson, for example, has managed to do it by redesigning the optical engine. S.M. Ram Prasad, senior business manager-Consumer Product Group, Epson India, explains, “Epson being the inventor of liquid crystal display (LCD) technology and pioneers in the space, we were able to redesign the optical engines to smaller sizes with no loss in light, using a better aperture ratio, with Epson LCD panels to achieve higher brightness.”

Acer, on the other hand, combines the finest portable and ultra-portable digital light processing (DLP) technology with micro-sized chipsets to create slimmer, lighter and fully programmable projectors. The company recently launched a model, the K10, that weighs just a little over half a kilogram. It uses a revolutionary LED technology that emits brightness of up to 100 ANSI lumens for clear and detailed images up to 152 cm (60 inches) diagonally. The use of LED technology ensures smaller footprints and design flexibility.



Epson EB-1910

One of the techniques that Canon uses to reduce the size of its micro-portable projectors is a free surface mirror within its optical box. “Canon is the only manufacturer to incorporate a free surface mirror within a projector’s optical box. The free surface mirror makes two important contributions to projector design: It helps minimise the number of parts used to process red light inside the optical box; and it contributes to a lighter and more compact body – an essential feature for micro-portable projector users,” says Nasir.

However, the deciding factor is not always ‘small size and bright and sharp images,’ though this might be the most widely used *mantra* for selection. Sometimes there is a trade-off of one or both factors, based on the model’s performance with respect to other parameters.

“There is often a trade-off between small size and image quality. A 5-7kg (12-15-pound) LCD offers a better combination of brightness and price, yet it is quite portable. The end image projection depends on the room brightness and screen size. For example, if a room is very well lit, such as with all fluorescents ‘on,’ one will need lots of lumens. If that room were to have only half of its fluorescents ‘on,’ one can get by with half the lumens. Rooms with minimal lighting will require a far less powerful projector. Simultaneously, the bigger the screen, the more lumens one will need. Therefore the size and brightness depend on various other factors,” explains Gautam Ghosh,

country manager, ViewSonic Technologies (India).

Plus, every consumer vertical will choose based on a varying set of parameters such as brightness, resolution, interfaces, weight and



NEC NP500WS

lamp life. While some segments such as businesses might focus on portability, others such as the education sector might consider price a more important factor. As a result, most companies offer generic as well as niche models. Canon, for example, has the XEED range for high-end applications such as simulation.

High-definition video is not just for TV

Gone are the days when projectors were used only to telecast slideshows. Thanks to the increasing performance and image quality of current projectors, they are now used to project high-definition movies, videoconferencing, games, simulations and what not. Another significant development that has led to the entry of projectors into the home entertainment and videoconferencing arenas is short-throw capability demonstrated by several brands including ViewSonic. This enables use of projectors even in small living rooms or conference rooms, without shadows, echoes and other such problems. Moreover, today’s projectors are compatible with several video formats.

“Considering the current developments in videoconferencing, we can see more of high-definition (HD) video signal communication. In sync, the industry is also following the current wide technology and manufacturing projectors suiting applications with 16:9 and 16:10 aspect ratio signals,” says Gupta of Actis Technologies.

Projectors have also started including CD/DVD drives and high-defini-